PocketPlate

An AI-Powered Budget Optimization and Multi-Restaurant Food Ordering Application for University Students

Project Proposal

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Abstract

University students often operate under strict financial constraints that significantly influence their daily food choices. While existing food delivery platforms provide convenience and variety, they lack intelligent budget-aware recommendation mechanisms tailored specifically for students. Most platforms allow price filtering but do not optimize meal selection based on available budget, nutritional value, ratings, or multi-restaurant combinations.

This project proposes the development of **PocketPlate**, an AI-powered budget optimization and multi-restaurant food ordering application designed specifically for university environments. The system will allow students to enter their available budget, preferences, and dietary requirements, after which it will intelligently recommend optimized meal combinations. Unlike traditional food delivery applications, PocketPlate will support cross-restaurant meal optimization and allow users to place multiple orders simultaneously.

The system will integrate artificial intelligence techniques such as constraint-based optimization and recommendation algorithms to maximize value within a defined budget. The application will directly connect students to partnered restaurants without managing delivery logistics, ensuring a scalable and cost-effective model.

Introduction

Food delivery applications have become an integral part of student life. Platforms such as Foodpanda and UberEats provide a wide range of restaurant options and delivery services. However, these systems do not provide intelligent budget-aware recommendations tailored to students with fixed allowances.

Students typically operate within daily or monthly financial limits and must make optimized food choices accordingly. Existing systems only allow price filtering rather than true budget optimization. Moreover, they do not allow intelligent cross-restaurant meal construction, such as selecting the best burger from one restaurant, the best fries from another, and dessert from a third vendor within a single optimized recommendation.

There is a clear need for a system that integrates budget constraints, optimization algorithms, and personalized recommendation models. PocketPlate aims to address this gap by developing a mobile application that intelligently suggests meals based on financial limitations while maximizing value, ratings, nutritional content, and user satisfaction. The system will also introduce simultaneous multi-restaurant order placement, providing flexibility not commonly available in current platforms.

Goals and Objectives

Our project’s objectives include:

- Designing and developing a mobile-based food ordering application for university students.
- Creating a centralized database for restaurants, menus, prices, ratings, and categories.
- Implementing a budget-constrained optimization engine to recommend meals within a user-defined budget.
- Applying algorithmic techniques (such as Knapsack optimization) to maximize value within budget constraints.
- Developing an AI-based intelligent meal builder capable of suggesting cross-restaurant meal combinations (e.g., burger from Restaurant A, fries from Restaurant B, dessert from Restaurant C).
- Enabling simultaneous placement of multiple restaurant orders in a single session.
- Designing a budget tracking module to monitor daily and monthly spending.
- Creating a student budgeting module that tracks daily and monthly food spending.
- Providing personalized recommendations based on user order history and preferences.
- Creating separate modules for students, restaurant partners, and administrators.
- Designing a user-friendly mobile application interface.
- Ensuring complete use of GitHub for version control and Jira for Agile sprint management throughout development.

Scope of the Project

Our project focuses on developing an AI-powered mobile application that connects university students directly with nearby restaurants and small food vendors. The system will provide intelligent recommendations and facilitate order placement.

The scope includes:

- User registration, authentication, and profile management.
- Restaurant partner registration and menu management.
- Budget input and preference selection by users.
- AI-based recommendation and optimization engine.
- Multi-restaurant cart and simultaneous order functionality.
- Spending analytics and budget tracking dashboard.
- Administrative control panel for system monitoring.

The system will not include:

- Management of delivery personnel or logistics operations.
- Real-world payment gateway implementation (payment may be simulated).
- Tax handling or legal compliance management.

The application will function as an intelligent recommendation and ordering platform rather than a full delivery management service.

Initial Study and Work Done So Far

An initial market study was conducted to analyze the features and limitations of existing food delivery platforms such as Foodpanda and UberEats. These platforms provide filtering and sorting mechanisms but do not implement budget-constrained optimization or cross-restaurant intelligent meal construction.

Research was conducted in the areas of recommendation systems and optimization algorithms. Studies related to the Knapsack problem were reviewed to understand budget allocation strategies under constraint conditions. Literature on hybrid recommendation systems and multi-criteria decision-making models was examined to explore personalization techniques.

Tutorials and technical documentation related to mobile application development, RESTful API design, database schema modeling, and Agile project management were studied to assess implementation feasibility. GitHub workflows for collaborative development and Jira-based sprint planning methodologies were also reviewed to ensure structured project execution.

The project will follow an Agile development approach, with tasks divided into sprints and tracked through Jira. All source code, updates, and documentation will be maintained through GitHub as mandated by course requirements.

References

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